

Software Requirements Specification

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Executive Summary

This report presents a comprehensive roadmap for the design and implementation of the Stock Management System at MININFRA. The system is being developed to automate manual inventory tracking processes, reduce human error, improve operational efficiency, and provide real-time visibility into stock levels and movement. The report also outlines stakeholder expectations, system roles, technical architecture, design requirements, and future improvement suggestions to guide the software development team in implementation.

Current inventory practices at MININFRA rely heavily on physical stock-cards and manual data entry, leading to inefficiencies, delays, and inaccuracies. Through direct consultation with the Logistics Department and system users, the following problems were identified:

- Inability to accurately track stock movement in real-time.
- Limited reporting capabilities for decision-making.
- Absence of digital records for audit and accountability.
- No user-based control or access permissions.

Stakeholder Vision: A centralized, digital, and real-time system that supports multiple user roles, generates insightful reports, and reduces inventory management workload.

1. Introduction

1.1. Purpose of the document

A software requirements specification (SRS) is a document that describes what the software will do and how it will be expected to perform. It also describes the functionality the product needs to fulfill all **Admin** and **User** needs.

2. User Requirements Specification

2.1. User Characteristics

The users include:

- **Institutions:**
- Head of Logistics Officer (Admin)
- Stock Keeper (User)

For this system, the **Admin** and **user** is required to know the basic usability of internet platforms as well as a very base level understanding of access, which hopefully will be facilitated by the software team through training.

2.2. User stories

2.2.0. Stock Keeper (User) Stories

Login: As a user, I would like to login on the platform, so that I can access it using The Registered Job E-mail, and corrected/Strong Password.

Logout: As a user, I would like to logout on the platform, so that I can exit it.

Forgot password: As a user, I would like to recover my forgotten password on the platform, so that I can access the system.

View Dashboard: As a user, I would like to view the User-dashboard for my processed requests by filtered criteria (Dashboard, Inventory-Management, Asset-Management, Reports, and Logout)

Dashboard management: As a user, I would like to view and Download a detailed list when I click on a dashboard item.

Download report for Inventory-Items: As a user, I would like to filter and download the list of Inventory-Item, so that I can get the customized report Based on the (Date, Item-Category, etc.....).

Download report for Asset-Items: As a user, I would like to filter and download the list of Asset-Item, so that I can get the customized report Based on the (Date, Item-Category, etc.....).

2.2.0.1. Specific Stock-Keeper (user) stories

2.2.0.1.1. Stock-Keeper (user) story

- **US1:** As a Stock-Keeper (User), I would Add Item quantities both in Asset-Management and Inventory-Management.
- **US2:** As a Stock-Keeper (User), I would Remove Item quantities both in Asset- Management and Inventory- Management.
- **US3:** As a Stock-Keeper (User), I would update/edit item quantities both in Asset-Item and Inventory-Item.
- **US4:** As a Stock-Keeper (User), I would view all details about item (Number, Item-Name, Description, Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Inventory-Management and (Number, Item-Name, Description, SKU(Code), Condition(Fair, Good, Very Good),Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Asset-Management.
- **US5:** As a User, I would like to receive the notification regarding the Item status so that it can report if the item is in Low-status (Stock Threshold).

2.2.1. Head of Logistic Officer (Admin) Stories

Login: As an admin, I would like to login on the platform, so that I can access it, using The Registered Job E-mail, and corrected/Strong Password.

Logout: As an admin, I would like to logout on the platform, so that I can exit it.

Password reset: As an admin, I would like to reset the password on the platform, so that I can have a secured one.

Forgot password: As an admin, I would like to recover my forgotten password on the platform, so that I can access the system.

View Dashboard: As an Admin, I would like to view the Admin-dashboard for my processed requests by filtered criteria (User-Dashboard, User-Management, Inventory-Management, Asset-Management, Reports, and Logout)

Dashboard management: As an Admin, I would like to access the User-Dashboard any Time to view and Download a detailed list.

Download report for Inventory-Items: As an Admin I can access the User-Dashboard, Then I would like to filter and download the list of Inventory-Item, so that I can get the customized report Based on the (Date, Item-Category, etc.....).

Download report for Asset-Items: As an Admin I can access the User-Dashboard, I would like to filter and download the list of Asset-Item, so that I can get the customized report Based on the (Date, Item-Category, etc.....).

Download report for Asset-Items: As an Admin I can access the User-Dashboard, I would like to filter and download the list of User-Management, so that I can get the customized report Based on the (Date is created, Full-Name, Job-Email, Telephone-Number and Password (is Hashed)).

2.2.1.1. Specific Head of Logistic Officer (Admin) stories

2.2.0.1.1. Head of Logistic Officer (Admin) Story

- **US1:** As a Head of Logistic Officer (Admin), I would Add Item (Number, Item-Name, Description, Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Inventory-Item and (Number, Item-Name, Description, SKU(Code), Condition(Fair, Good,

Very Good),Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Asset-Item.

- **US2:** As a Head of Logistic Officer (Admin), I would update/edit item (Number, Item-Name, Description, Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Inventory-Item and (Number, Item-Name, Description, SKU(Code), Condition(Fair, Good, Very Good),Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Asset-Item.
- **US3:** As an Head of Logistic Officer (Admin), I would view all details about item (Number, Item-Name, Description, Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Inventory-Item and (Number, Item-Name, Description, SKU(Code), Condition(Fair, Good, Very Good),Quantities-in, Quantities-out, Balance in Quantities, Unit-Price, Total-price) For Asset-Item.
- **US4:** As a Head of Logistic Officer (Admin), I would like to receive the notification regarding the Item status so that it can report if the item is in Low-status (Stock-Treshold) through in User-Dashboard.
- **US5:** As a Head of Logistic Officer (Admin) , I would like To add The new User with (Full-Name, Job-Email, Telephone-Number and Password)
- **US6:** As a Head of Logistic Officer (Admin), I would like To Edit/Update The new User with (Full-Name, Job-Email, Telephone-Number and Password)
- **US7:** As a Head of Logistic Officer (Admin), I would like To Disable (Revoke) or Block the User any time.
- **US8:** As a Head of Logistic Officer (Admin), I would like To Enable or Grant the User any time.
- **US8:** As a Head of Logistic Officer (Admin), I would like To Assign the inventory-management and Asset-management to the Stock-Keeper (User)

3. System Features and Requirements

3.1. Functional Requirements

In reference to elicited user stories above, below are the actual user requirements per user category.

3.1.1. Use Case Analysis and Elicitation

3.1.1.1. General use cases

Use case 1: User/Admin Login

User Story 1: As a user/admin, I would like to login on the platform, so that I can access it.

Initiator	Every system user
Pre-condition	The user must have an active account on platform
Use case elicitation	• User elects to log in the platform • System displays a login form • User fills the login form with a username (a username can be an Job Email and a password); then submits the form • System validates the login credentials and responds with a successful login, and the use case ends.
Alternative cases	A1: If the username and password are not valid the login must be aborted, and the user must be told that the username or password is not valid.
Business rules	Rule 1: The system should allow login to credentials that match with the registered ones.

Use case 2: User Logout

User story 1: As a user/admin, I would like to logout on the platform, so that I can exit it.

Initiator	Every system user
Pre-condition	The user must be logged in to the platform
Use case elicitation	• User elects to log out the platform • System provides a logout button/Icon • User clicks the logout button /Icon • System displays a confirmation popup (Are you sure you want to exit? Yes or No) • User clicks the option Yes • System redirects to the login form
Alternative cases	A1. If the user clicks on option No, The session remains Active and the user is redirected where he was.
Business rules	Rule 1: The user session must be active.

Use case 3: Forgot password

User story 1: As a user or Admin, I would like to recover my forgotten password on the platform, so that I can access the system.

Initiator	Every system user
Pre-condition	The user must remember his registered Job-email
Use case elicitation	• User elects to restore forgotten password on the platform • System displays a forget password form • User fills the form with the email; then submits the form • System sends the reset password link to the registered email.

	• System sends an SMS containing the reset password link to the registered phone number • User clicks the link received via email • System displays a reset password form • User fills the reset password form with new password; then submits the form • System validates the reset password process and responds with a positive response, and the use case ends.
Alternative cases	A1: If the filled password is not strong, the reset password process must be aborted/Cancelled/Revoked, and the user must be told to provide a strong password.
1. Business rules	Rule 1: The system should only allow strong passwords consisting of at least 8 characters, one special character, one upper case letter, and one digit. Rule 2: The system should only allow a registered Job Email Rule 3: The system should provide the reset password with 2 parameters: A unique user id and a generated token that expires once the link is clicked; Rule 4: The system should provide the expiration time for the reset password link even if it has not been used like in 24 hours only. Rule 5: The system should avoid previously used passwords while resetting Rule 6: It is the responsibility of the user/ Admin to keep his/her email safe from unauthorized access.

Use case 2: User/Admin Logout

User story 1: As a user/admin, I would like to logout on the platform, so that I can exit it.

Initiator	Every system user
Pre-condition	The user must be logged in to the platform
Use case elicitation	• User elects to log out the platform • System provides a logout button/Icon • User clicks the logout button /Icon • System displays a confirmation popup (Are you sure you want to exit? Yes or No) • User clicks the option Yes • System redirects to the login form
Alternative cases	A1. If the user clicks on option No, The session remains Active and the user is redirected where he was.
Business rules	Rule 1: The user session must be active.

Use case 4: Admin

User story 1: As an Admin, I would like to access the Admin-Dashboard on the platform.

Initiator	Head of Logistics officer (Admin)
Pre-conditions	Admin must be logged in to the platform
Use case elicitation	• Admin can access the User-Dashboard and able to show and navigate all activities that done by the Stock-Keeper(User) • The system displays the Pages and select to go to the User-Dashboard, User-Management, Asset-

Management, Inventory-Management, Reports and Logout

1. Admin selects a User-Dashboard
The system directly go to the User-Dashboard
2. Admin selects a User-Management and see all user are in the system, add, edit/update, and other activities done by admin in the User-Management
3. Admin selects a Asset-Management and see all Asset are available in the system, add, edit/update, and other activities done by admin in the Asset-Management
4. Admin selects an Inventory-Management and see all Inventory are available in the system, add, edit/update, and other activities done by admin in the Inventory-Management
5. Admin can be able to generate the reports for User Management through the Admin-Dashboard and be able to generate the Asset-management and Inventory-Management Through The User-Dashboard
6. Admin can be able to logout in the system when they click on the logout button he/she can receive the Pop-up message inform him/her to logout when he/she can click yes the system directly go to the login page this means the logout session is successful

	while he click on NO the system remains on the Admin Dashboard
Alternative cases	The system validates the request and responds positively, and the use case ends.

Use case 5: User

User story 1: As an Store-Keeper (User), I would like to access the User-Dashboard on the platform.

Initiator	Store-Keeper (User)
Pre-conditions	User must be logged in to the platform
Use case elicitation	- The system displays the Pages and select go to the Dashboard, Asset-Management, Inventory-Management, Reports and Logout 1. User selects a Dashboard The system directly go to the Dashboard and show the Total items, Items in inventory-stock, Items in Asset-stock, Low Stock (Stock Treshold) and this its updated for every Transaction it done. 2. User selects an Asset-Management and see all Asset are available in the system, and it's able to make the transaction of stock-out and stock-in for this case the user are able to deal with Quantity-in and Quantity-out only other things about the asset are fixed by the Admin after any transaction the system it directly

	calculate the balance are remain in the stock and the value of it. 3. User selects an Inventory-Management and see all inventory are available in the system, and it's able to make the transaction of stock-out and stock-in for this case the user are able to deal with Quantity-in and Quantity-out only other things about the inventory are fixed by the Admin, after any transaction the system it directly calculate the balance are remain in the stock and the value of it. 4. User can be able to generate the reports for Inventory-Management and Asset-Management based on their needs 5. User can be able to logout in the system when they click on the logout button he/she can receive the Pop-up message inform him/her to logout when he/she can click yes the system directly go to the login page this means the logout session is successful while he/she click on NO the system remains on the User Dashboard
Alternative cases	The system validates the request and responds positively, and the use case ends.